

PRACTICAL WEEK 1

Task 1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

- a) Write a C code to create a process using fork() system call.
- b) Read the user input as shell command.
- c) Use a system call exec() to execute the shell command.
- d) Use a wait() system call for waiting a parent process.
- e) Display the PIDs of parent and child processes.

1) Create a file

```
Radhakrishna@04:~# nano main.c
```

Then press enter to open command

```
Radhakrishna@04:~# nano main.c
Radhakrishna@04:~# gcc main.c -o main
```

```
GNU nano 7.2
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    char command[100];

    printf("Enter Linux command: ");
    fgets(command, sizeof(command), stdin);

    // Remove newline character
    command[strcspn(command, "\n")] = '\0';

    pid_t pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }
}
```

```

else if (pid == 0) {
    // Child Process
    printf("\nChild Process\n");
    printf("Child PID: %d\n", getpid());

    execl("/bin/sh", "sh", "-c", command, NULL);

    // Executes only if execl fails
    perror("Execution failed");
    exit(1);
}

else {
    // Parent Process
    printf("\nParent Process\n");

```

```

}
else {
    // Parent Process
    printf("\nParent Process\n");
    printf("Parent PID: %d\n", getpid());

    wait(NULL);

    printf("Child process completed.\n");
}

return 0;
}

```

- 1) To save press control O
- 2) To come back to shell press control X
- 3) Then for compilation write a command
gcc filename.c -o filename

Then press enter then to view a output write a command
./filename

```

Radhakrishna@04:~# nano main.c
Radhakrishna@04:~# gcc main.c -o main
Radhakrishna@04:~# ./main
Enter Linux command: ls

Parent Process
Parent PID: 470

Child Process
Child PID: 473
sh: 1: ls: not found
Child process completed.

```

Then you will be asking to enter linux command to view the output

In ENTER LINUX COMMAND: enter either "ls" command (or) "pwd" command to see the output of your code

```
Radhakrishna@04:~# nano main.c
Radhakrishna@04:~# gcc main.c -o main
Radhakrishna@04:~# ./main
Enter Linux command: ls

Parent Process
Parent PID: 470

Child Process
Child PID: 473
sh: 1: ls: not found
Child process completed.
```

Task-2: Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```
Radhakrishna@04:~# uname Linux
uname: extra operand 'Linux'
Try 'uname --help' for more information.
Radhakrishna@04:~# nano main.c
Radhakrishna@04:~# gcc main.c -o main
Radhakrishna@04:~# ./main
Enter Linux command: uname

Parent Process

Child Process
Parent PID: 481
Child PID: 482
Linux
Child process completed.
```

```

Radhakrishna@04:~# nano main.c
Radhakrishna@04:~# gcc main.c -o main
Radhakrishna@04:~# ./main
Enter Linux command: lscpu

Parent Process
Parent PID: 491

Child Process
Child PID: 492
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:                39 bits physical, 48 bits virtual
Byte Order:                   Little Endian
CPU(s):                       8
On-line CPU(s) list:         0-7
Vendor ID:                    GenuineIntel
Model name:                   11th Gen Intel(R) Core(TM) i7-1185G
                             7 @ 3.00GHz
CPU family:                   6
Model:                        140
Thread(s) per core:          2
Core(s) per socket:          4
Socket(s):                    1
Stepping:                     1
BogoMIPS:                     3609.60
Flags:                         fpu vme de pse tsc msr pae mce cx8
                             apic sep mtrr pge mca cmov pat pse3
                             6 clflush mmx fxsr sse sse2 ss ht s
                             yscall nx pdpe1gb rdtscp lm constan
                             t_tsc arch_perfmon rep_good nopl xt
                             opology tsc_reliable nonstop_tsc cp
                             uid tsc_known_freq pni pclmulqdq vm
                             x ssse3 fma cx16 pdcm pcid sse4_1 s
                             se4_2 x2apic movbe popcnt tsc_deadl
                             ine_timer aes xsave avx f16c rdrand
                             hypervisor lahf_lm abm 3dnowprefet
                             ch ssbd ibrs ibpb stibp ibrs_enhanc
                             ed tpr_shadow ept vpid ept_ad fsgsb
                             ase tsc_adjust bml avx2 smep bmi2
                             erms invpcid avx512f avx512dq rdsee
                             d adx smap avx512ifma clflushopt cl
                             mb avx512cd sha_ni avx512bw avx512v
                             l xsaveopt xsavec xgetbv1 xsaves vn
                             mi avx512vbmi umip avx512_vbmi2 gfn
                             i vaes vpclmulqdq avx512_vnni avx51
                             2_bitalg avx512_vpopcntdq rdpid mov
                             dir64b fsrm avx512_vp2inter
                             sect md_clear ibt flush_lld arch_ca
                             pabilities

Virtualization features:
Virtualization:               VT-x
Hypervisor vendor:            Microsoft
Virtualization type:          full
Caches (sum of all):
L1d:                           192 KiB (4 instances)
L1i:                           128 KiB (4 instances)
L2:                             5 MiB (4 instances)
L3:                             12 MiB (1 instance)
NUMA:
NUMA node(s):                  1
NUMA node0 CPU(s):            0-7
Vulnerabilities:
Gather data sampling:          Unknown: Dependent on hypervisor st
atus

```

```

Vulnerabilities:
Gather data sampling:          Not affected
Ghostwrite:                    Not affected
Indirect target selection:     Not affected
Itlb multihit:                 Not affected
L1tf:                           Not affected
Mds:                           Not affected
Meltdown:                      Not affected
Mmio stale data:               Not affected
Old microcode:                 Not affected
Reg file data sampling:        Vulnerable: No microcode
Retbleed:                      Mitigation; Enhanced IBRS
Spec rstack overflow:          Not affected
Spec store bypass:             Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:                    Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:                    Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-IBRS SW sequence; BHI BHI_DIS_S
Srbds:                         Not affected
Tsa:                           Not affected
Tsx async abort:               Not affected
Vmscape:                       Not affected

```

```
Child process completed.
Radhakrishna@04:~# nano main.c
Radhakrishna@04:~#
Radhakrishna@04:~# gcc main.c -o main
Radhakrishna@04:~# ./main
Enter Linux command: ps
```

```
Parent Process
Parent PID: 502
```

```
Child Process
Child PID: 503
  PID TTY          TIME CMD
  444 pts/0        00:00:00 bash
  502 pts/0        00:00:00 main
  503 pts/0        00:00:00 sh
  504 pts/0        00:00:00 ps
Child process completed.
```

```
Radhakrishna@04:~# ./main
Enter Linux command: top
```

```
Parent Process
Parent PID: 505
```

```
Child Process
Child PID: 506
top - 04:03:12 up 7 min, 1 user, load average: 0.00, 0.01, 0.00
Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7635.6 total, 6473.2 free, 497.6 used, 814.7 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7138.0 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	21732	13196	9852	S	0.0	0.2	0:00.69	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
7	root	20	0	3196	2128	2008	S	0.0	0.0	0:00.00	init
56	root	19	-1	50444	16292	15204	S	0.0	0.2	0:00.18	systemd-journal
101	root	20	0	25284	6740	5200	S	0.0	0.1	0:00.19	systemd-udev
147	systemd+	20	0	21344	13276	11148	S	0.0	0.2	0:00.13	systemd-resolve
153	systemd+	20	0	91036	8060	7088	S	0.0	0.1	0:00.09	systemd-timesyn
168	root	20	0	4244	2744	2496	S	0.0	0.0	0:00.00	cron
169	message+	20	0	9596	5404	4720	S	0.0	0.1	0:00.09	dbus-daemon
177	root	20	0	17980	8948	7912	S	0.0	0.1	0:00.06	systemd-logind
179	root	20	0	1755848	13336	11144	S	0.0	0.2	0:00.11	wsl-pro-service
189	syslog	20	0	222516	6064	4668	S	0.0	0.1	0:00.08	rsyslogd
192	root	20	0	3124	2036	1900	S	0.0	0.0	0:00.00	agetty
214	root	20	0	107032	23148	13692	S	0.0	0.3	0:00.16	unattended-upgr
304	root	20	0	6672	4564	3796	S	0.0	0.1	0:00.01	login
349	root	20	0	20120	11384	9452	S	0.0	0.1	0:00.10	systemd
350	root	20	0	21168	3496	1784	S	0.0	0.0	0:00.00	(sd-pam)
374	root	20	0	6080	5296	3664	S	0.0	0.1	0:00.02	bash
440	root	20	0	3188	1116	980	S	0.0	0.0	0:00.00	SessionLeader
442	root	20	0	3204	1256	1108	S	0.0	0.0	0:00.05	Relay(444)
444	root	20	0	6080	5376	3692	S	0.0	0.1	0:00.06	bash
505	root	20	0	2688	1800	1700	S	0.0	0.0	0:00.00	main
506	root	20	0	2808	1908	1808	S	0.0	0.0	0:00.00	sh
507	root	20	0	9436	5836	3564	R	0.0	0.1	0:00.02	top

